Chapter 25 Problem Set

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1a. The equivalence class $[1,1]$ represents the number 0. So $[1,1]+[a,b]=[a,b]$ essentially means that adding 0 to a number results in the same number.

1b. Since $[1,1]$ represents 0, it says that 0 times a number is 0.

1c. This statement says that addition is commutative.

1d. This statement is equivalent to $x(y+z)=xy+xz$, which is the distributive property.

2. First, we need to prove that $(a,b)\equiv(a,b)$. This means that $ab=ab$, which is true. So $\equiv$ is reflexive. Now, we assume that $(a,b)\equiv(c,d)$, or $ad=bc$. That means $(c,d)\equiv(a,b)$ holds if $cb=da$, which we know is true. Finally, we assume that $(a,b)\equiv(c,d)$ and $(c,d)\equiv(e,f)$. This means that $ad=bc$ and $cf=de$. So $a=bc/d$ and $f=de/c$. Multiplying those equations, we get $af=bcde/cd=be$, which means $(a,b)\equiv(e,f)$, so $\equiv$ is transitive.

3. Since $a\leq b$, there exists $x\in\mathbb{N}$ such that $a+x=b$. Also, since $c\leq d$, there exists $y\in\mathbb{N}$ such that $c+y=d$. Adding the two equations, we get $a+c+(x+y)=b+d$. We know that $x+y\in\mathbb{N}$, so $a+c\leq b+d$.

4. By the definition, $ab>0$ if there doesn't exist $x\in\mathbb{N}\cup 0$ such that $ab+x=0$. Since $a,b<0$, $ab=(-1c)(-1d)=cd$ for $c,d\in\mathbb{N}$. Therefore, if $x\in\mathbb{N}\cup 0$ exists such that $ab+x=0$, then $x=-1\cdot cd$, which is impossible. So $ab>0$.

5. We know that $x+y=\langle x_1,x_2\rangle+\langle y_1,y_2\rangle=\langle 0,1\rangle$, so $\langle x_1,x_2\rangle=\langle 0,1\rangle-\langle y_1,y_2\rangle=\langle -y_1,y_2\rangle=\langle -1,1\rangle\langle y_1,y_2\rangle=-y$. Since we isolated x using reversible steps, we know that $-y$ is the only value that works.

6. Let $x=\langle x_1,x_2\rangle$ and $y=\langle y_1,y_2\rangle$. Then $\langle x_1,x_2\rangle\langle y_1,y_2\rangle=\langle 1,1\rangle$, so $\langle x_1,x_2\rangle=\langle 1,1\rangle/\langle y_1,y_2\rangle=\langle y_2,y_1\rangle$. So $x_1=y_2$ and $x_2=y_1$. This is the only solution because we isolated x using reversible steps.

7a. For $a,b,c,d\in\mathbb{Z}$, let's say $(a,b)\sim(c,d)$ if and only if $a2^d=c2^b$.

We know $(a,b)\sim(a,b)$ iff $a2^b=a2^b$, which is true. So $\sim$ is reflexive.

If $(a,b) \sim (c,d)$, then $a2^d = c2^b$, or $c2^b = a2^d$, which means $(c,d) \sim (a,b)$. So $\sim$ is symmetric.

Finally, if $(a,b) \sim (c,d)$ and $(c,d) \sim (e,f)$, then $a2^d = c2^b$ and $c2^f = e2^d$. Reversing the second equation and dividing it from the first, we get $\frac{a}{e} = \frac{2^b}{2^f}$, or $(a,b) \sim (e,f)$. So $\sim$ is transitive.

7b. Let's say $(a,b) + (c,d) = (a2^d + c2^b, b+d)$.

To show that this is well-defined, we must prove, given $(a_1,b_1) = (a_2,b_2)$ and $(c_1,d_1) = (c_2,d_2)$, that $(a_1+c_1, b_1+d_1) = (a_2+c_2, b_2+d_2)$. This means we have to check that $(a_1+c_1)2^{b_2+d_2} = (a_2+c_2)2^{b_1+d_1}$ given $a_1 2^{b_2} = a_2 2^{b_1}$ and $c_1 2^{d_2} = c_2 2^{d_1}$.

First, we multiply both sides of $a_1 2^{b_2} = a_2 2^{b_1}$ by 2^{d_2} and multiply both sides of $c_1 2^{d_2} = c_2 2^{d_1}$ by 2^{b_2} to get $a_1^{b_2+d_2} = a_2 2^{b_1+d_1}$ and $c_1^{b_2+d_2} = c_2 2^{b_1+d_1}$. Now, adding the two equations, we get $a_1 2^{b_2+d_2} + c_1 2^{b_1+d_2} = a_2 2^{b_1+d_2} + c_2 2^{b_1+d_1}$. Rearranging terms, substituting, and simplifying, we get $(a_1+c_1)2^{b_2+d_2} = (a_2+c_2)2^{b_1+d_1}$, as desired.

Now, let's say $(a,b) - (c,d) = (a2^d - c2^b, b+d)$. Similar to before, we must prove that $(a_1-c_1)2^{b_2+d_2} = (a_2-c_2)2^{b_1+d_1}$ given $a_1 2^{b_2} = a_2 2^{b_1}$ and $c_1 2^{d_2} = c_2 2^{d_1}$.

First, multiply both sides of $a_1 2^{b_2} = a_2 2^{b_1}$ by 2^{d_2} and multiply both sides of $c_1 2^{d_2} = c_2 2^{d_1}$ by 2^{b_2} to get $a_1^{b_2+d_2} = a_2 2^{b_1+d_1}$ and $c_1^{b_2+d_2} = c_2 2^{b_1+d_1}$. Subtracting the two equations gives us $a_1 2^{b_2+d_2} - c_1 2^{b_1+d_1} = a_2 2^{b_1+d_2} - c_2 2^{b_1+d_1}$. Simplifying this gives us $(a_1-c_1)2^{b_2+d_2} = (a_2-c_2)2^{b_1+d_1}$, as desired.

Define $(a,b) \times (c,d) = (ac, b+d)$. Given $a_1 2^{b_2} = a_2 2^{b_1}$ and $c_1 2^{d_2} = c_2 2^{d_1}$, we must prove that $(a_1 c_1)2^{b_2+d_2} = (a_2 c_2)2^{b_1+d_1}$.

We can rewrite the given equations as $a_1 = a_2 2^{b_1-b_2}$ and $c_1 = c_2 2^{d_1-d_2}$. Multiplying the equations gives us $a_1 c_1 = a_2 c_2 2^{b_1-b_2+d_1-d_2}$. Multiplying both sides by $2^{b_2+d_2}$ gives $a_1 c_1 2^{b_2+d_2} = a_2 c_2 2^{b_1+d_1}$, as desired.

To define division, we might try $(a,b)/(c,d) = (a2^d, c\log_2 b)$. But logarithms don't consistently produce integers. If, instead, we try to define division as $(a,b)/(c,d) = (a/c, b-d)$ where $c \neq 0$, we run into a problem, since $\frac{a}{c}$ isn't consistently an integer. So it seems like we cannot define division of any two dyadic rationals straight from $\mathbb{Z}$.

8. Letting a, b, c, and d be rational in the construction of $\mathbb{Q}$ gives us $(a,b) \sim (c,d)$ if $ad = bc$, or $\frac{a}{b} = \frac{c}{d}$. Dividing two rationals gives us another rational, so this construction seems to just give us $\mathbb{Q}$ back.

9. Let a, b, c, and d be complex. Then $(a, b) \sim (c, d)$ if $ad = bc$, or $\frac{a}{b} = \frac{c}{d}$. If we let $a = a_1 + a_2 i$ and $b = b_1 + b_2 i$, then $\frac{a}{b} = \frac{a_1 + a_2 i}{b_1 + b_2 i} = \frac{a_1 b_1 + a_2 b_2}{b_1^2 + b_2^2} + \frac{a_2 b_1 - a_1 b_2}{b_1^2 + b_2^2} i$, which is essentially a complex number with rational real and imaginary parts. So the set described in the problem seems to represent the set of complex numbers with rational real and imaginary parts.